



COMSATS University Islamabad, Wah Campus
Civil Engineering Department
Assignment - 1

Program: CEE 104
Course: Civil Engineering Materials
Due Date: 16 Feb 2026

Total Marks: 10
Instructor: Dr. Adnan Nawaz

[C2, CLO-1, PLO1], [C2, CLO-3, PLO6]

1. The Objective

Civil Engineering is moving from the "Portland Era" (Cement 1.0) toward a "Smart & Green Era" (Cement 2.0/3.0). As future engineers, you must be able to compare the technical specifications of local materials with global innovations to build sustainable infrastructure. This assignment challenges you to move beyond local manufacturing and benchmark **Pakistan's Cement Industry** against **Global Net-Zero Leaders**. You will analyze how companies are balancing production with environmental responsibility.

2. The Task

Your group must produce a **Technical Dossier** comparing **four Companies**:

- **2 Local Companies:** (e.g., Lucky Cement, Bestway, DG Khan, etc.).
- **2 International Companies:** (e.g., Holcim, Heidelberg Materials, or CEMEX, etc.).

3. Technical Requirements

Your dossier must analyze these companies through the lens of the **"3 Versions of Cement"**:

Part A: The Standard (Cement 1.0) – CLO 1

- **Manufacturing:** Identify if the plants use the **Dry or Wet process**.
- **Specifications:** Identify the **ASTM or EN Standards** mentioned on their packaging (e.g., ASTM C150 for OPC, or EN 197-1).
- **Physical Properties:** Compare the setting times and the 3-day, 7-day, and 28-day compressive strengths if available in their technical data sheets.
- **Chemistry:** Identify the percentage of Clinker used in their primary products.
- **Grade Analysis:** Identify the cement grades produced (e.g., 43 Grade vs 53 Grade).
- **Any other details:** Any other technically relevant information provided on the packaging

Part B: The Sustainability Component: The Transition (Cement 2.0) – CLO 3

- **Carbon Strategy:** Analyze how these companies handle emissions.
- **Sustainable Binders:** Research if the companies are using **Blended Cements** (Fly Ash, Slag, or Calcined Clay/LC3).
- **Carbon Mitigation:** Investigate their use of **Waste Heat Recovery (WHR)** or **Carbon Credits/Offsetting**. How do the 2 international companies compare to our local ones in carbon-neutral goals?

Part C: The Future (Cement 3.0) – Futuristic Element – CLO 3

- **Innovation Spotlight:** Select **one** "Future Tech" from your international companies and explain how it works (e.g., **Self-healing concrete**, **Carbon Capture (CCUS)**, or **Concrete that absorbs CO₂**).
- **Feasibility:** Briefly state if this "3.0" technology could be implemented in the Pakistani construction market in the next decade. Support your feasibility argument with at least one local constraint (e.g., cost, codes, materials availability, or industry readiness) in Pakistan.

Students should reference at least one Technical Data Sheet (TDS) or Sustainability Report per company. Provide a reference list at the end of the dossier.

4. Deliverables (Choose One)

Students are encouraged to use digital-first formats:

1. **Digital Benchmarking Dashboard:** A high-impact visual comparison (using Canva/Visme).
2. **The "Consultant's Presentation":** A professional slide deck with embedded data charts.
3. **Video Industry Briefing:** A 5-minute "news-style" video explaining the shift from 1.0 to 3.0.

5. AI Disclaimer & Collaboration Map (Template)

You may use AI (Gemini/ChatGPT, etc.) to summarize international technical reports. However, any numerical data obtained through AI assistance must be cross-checked with standard codes, technical data sheets, or official company publications. Furthermore, you must include a similar table on your final page:

Task Component	AI Tool Used	AI Contribution	Human Input & Verification
Technical Data	e.g., ChatGPT, Perplexity	e.g., Summarized ASTM C150 specs/Summarized the Cement company's 2025 report	Checked against the course textbook/Cross-checked stats with the company website
Comparison	e.g., Gemini, ChatGPT	Drafted the 1.0 vs 2.0 table	Verified local Pakistani plant data
Formatting/ Visuals/Script	e.g., Canva AI	Generated layout theme/Generated icons or script for video	Edited for engineering terminology

Standard Disclosure Statement:

"We acknowledge the use of AI to assist in [researching/formatting/summarizing] this assignment. We take full responsibility for the technical accuracy of the engineering data provided."

Student Checklist for Success

- ☐ Analyzed 2 Local + 2 International companies.
- ☐ Identified the **Standard Specs** (ASTM/EN) for all 4.
- ☐ Explained the difference between **Cement 1.0** and **2.0** for your chosen firms.
- ☐ Included one "**Cement 3.0**" futuristic technology.
- ☐ Attached the signed **AI Collaboration Map**.